

第九讲：图像识别



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2026-3-31



Object recognition 目标识别

- Typically, this is a classification problem 相当于分类问题

dog



?

cat



rabbit



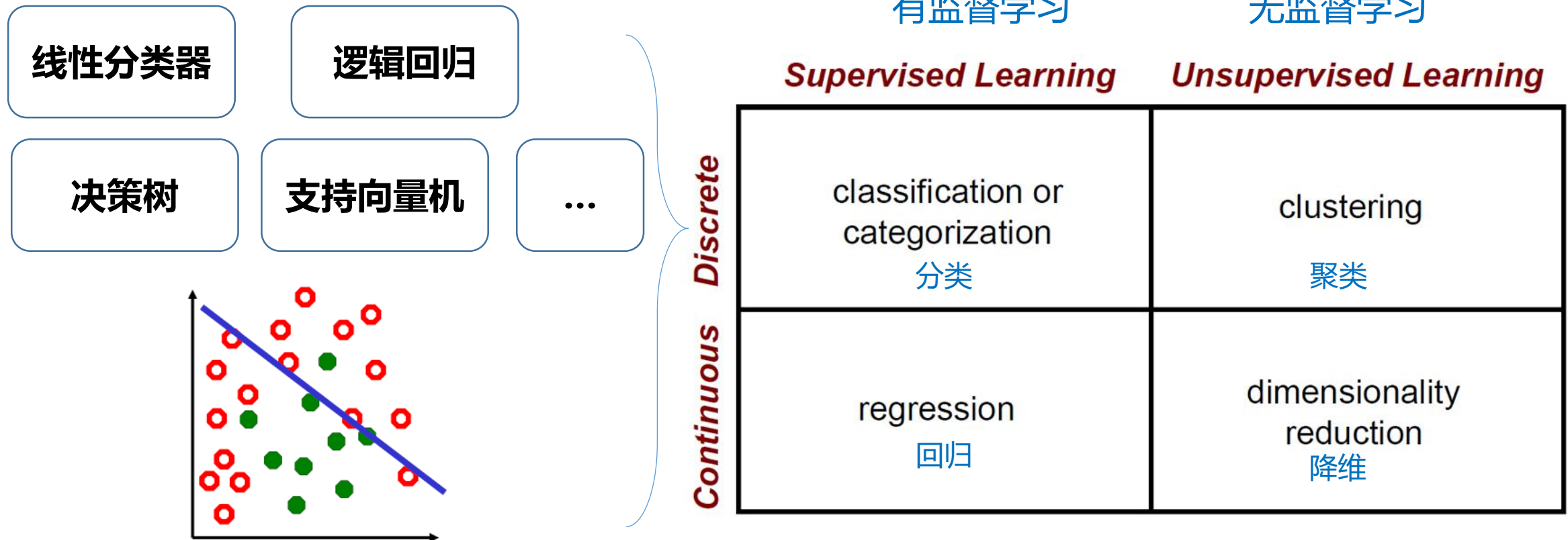
Content



- Supervised learning 监督学习
 - Linear and non-linear classifiers
 - Underfitting and overfitting
 - Model evaluation
- Bag of features models 特征袋模型
 - Bag of words
 - Spatial pyramid representation
 - Applications

Machine learning problems

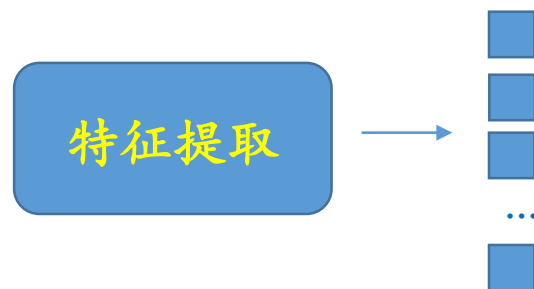
- Basic categories: supervised and unsupervised learning



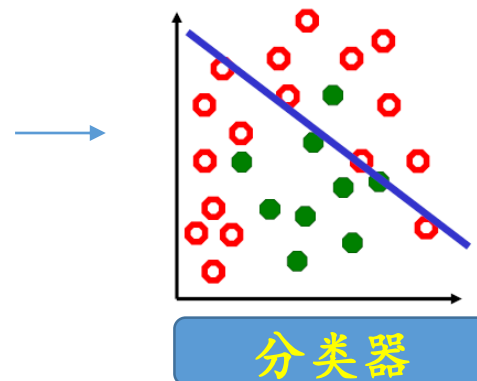
图像识别基本框架



输入图像



特征向量



鲈鱼?

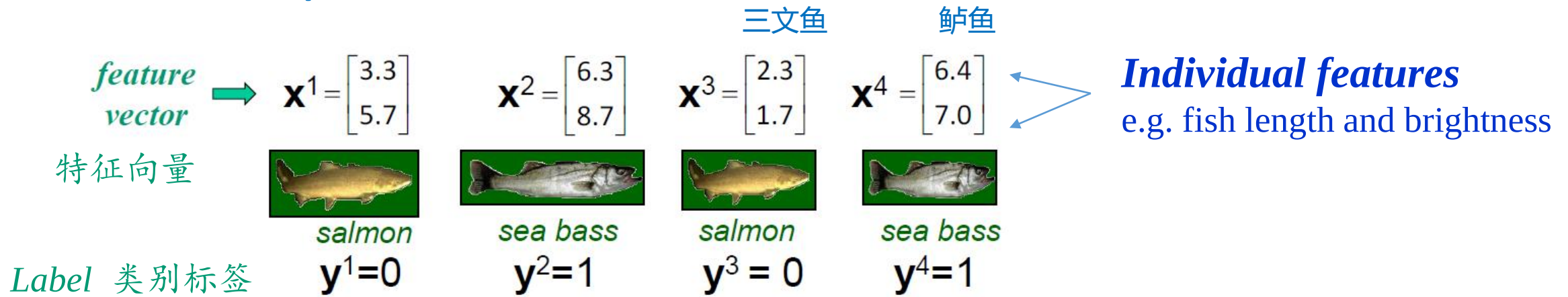
是
否

识别结果

Supervised learning 有监督学习



- Have training examples with corresponding outputs/labels
- For **example: fish classification - salmon or sea bass?**



Each example should be represented by *feature vector* $\mathbf{x}^i$

- data may be given in vector form from the start
- if not, for each example i , extract useful features, put them as a vector

y^i is the output (*label* or target) for example $\mathbf{x}^i$

Supervised learning 有监督学习

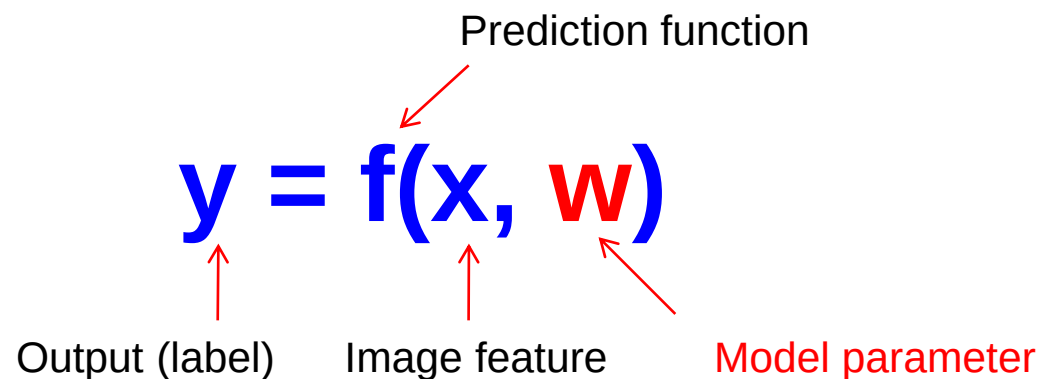


- Formulation of supervised learning

$$y = f(x, w)$$

Prediction function

Output (label) Image feature Model parameter

A diagram showing the equation y = f(x, w). Red arrows point from the labels 'Output (label)', 'Image feature', and 'Model parameter' to the variables y, x, and w respectively. Another red arrow points from the label 'Prediction function' to the function symbol f.

训练过程

Training: Given a *training set* of labeled examples:

$$\{(x_1, y_1), \dots, (x_N, y_N)\}$$

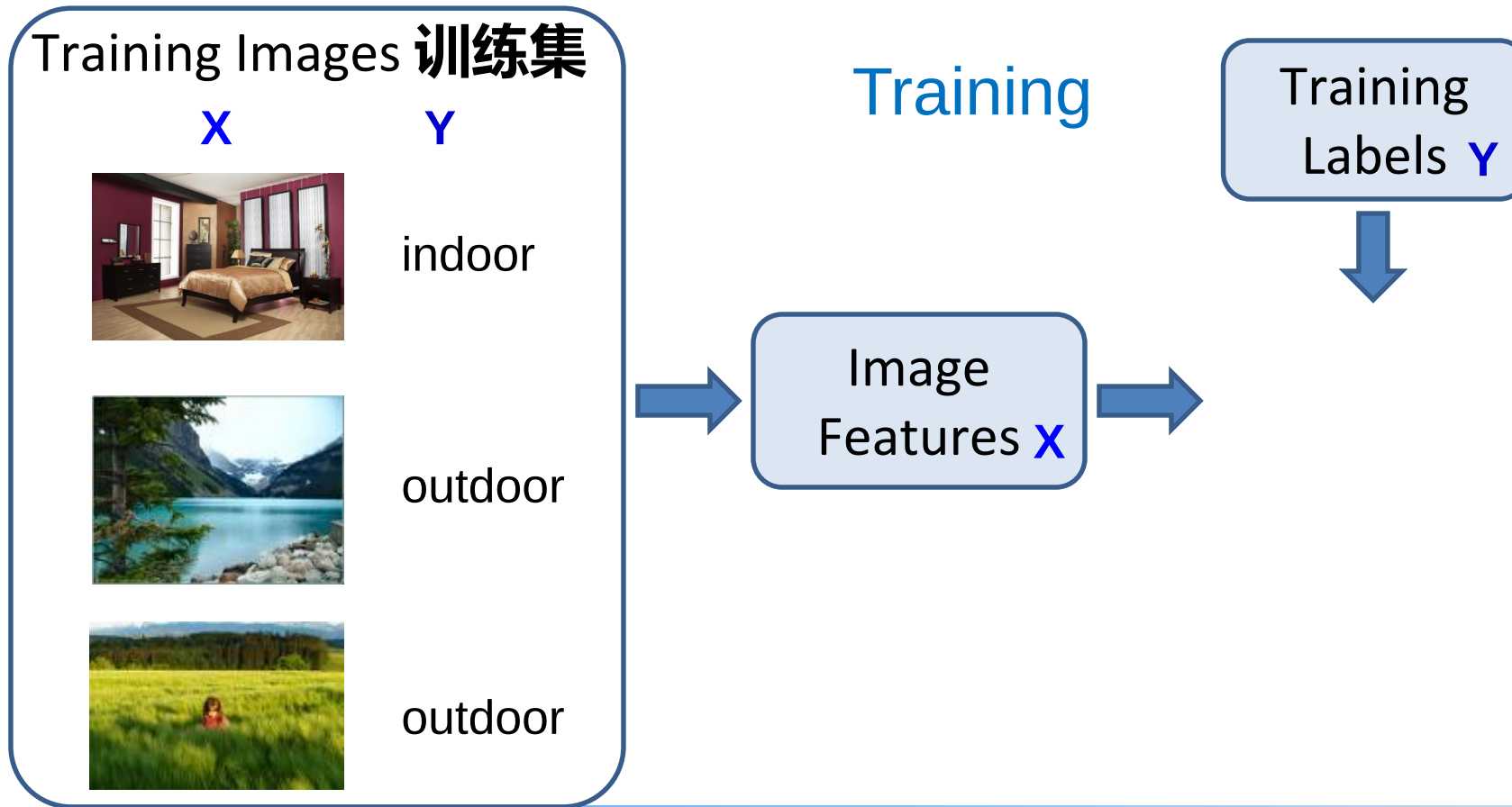
Estimate the prediction function f by minimizing the prediction error on the training set.

测试

Testing: Apply f to a unseen *test example* x and output the predicted value $y = f(x)$ to *classify* x .

Supervised image classification

- We start with a training set with labels



Supervised image classification

- We start with a training set with labels

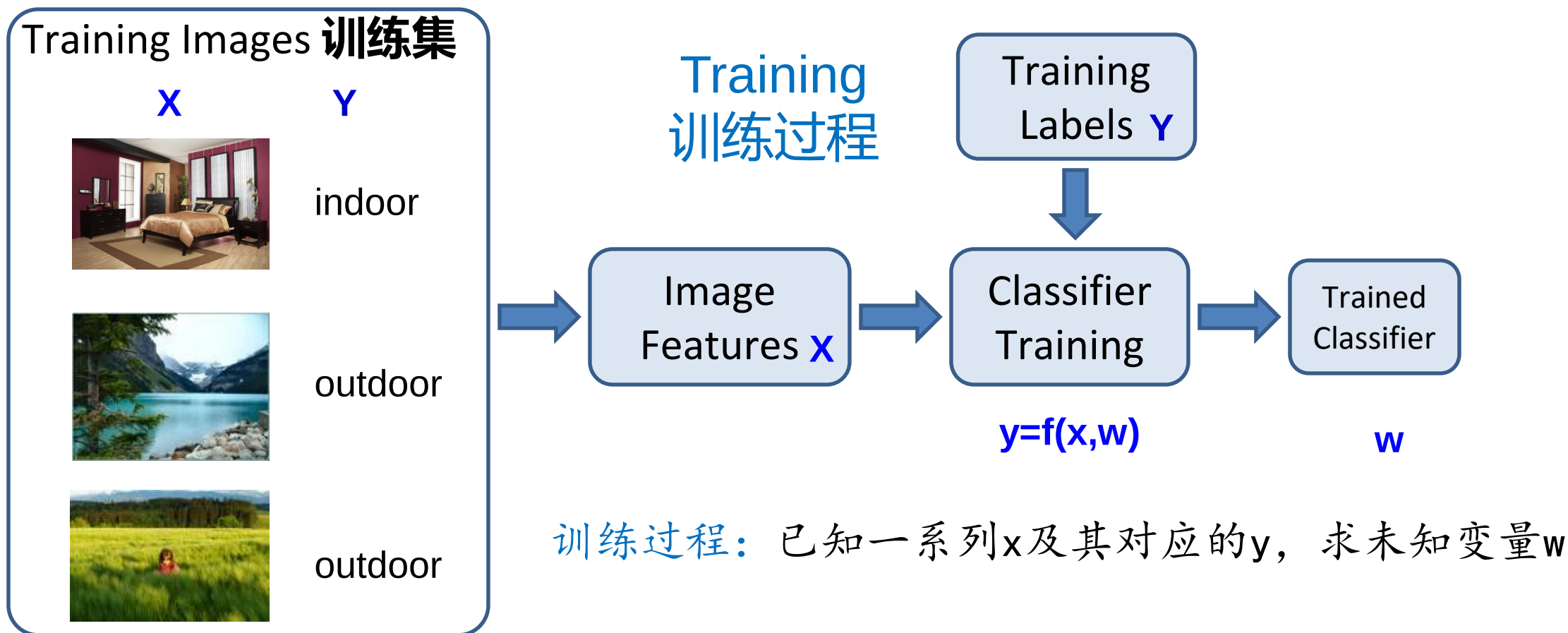
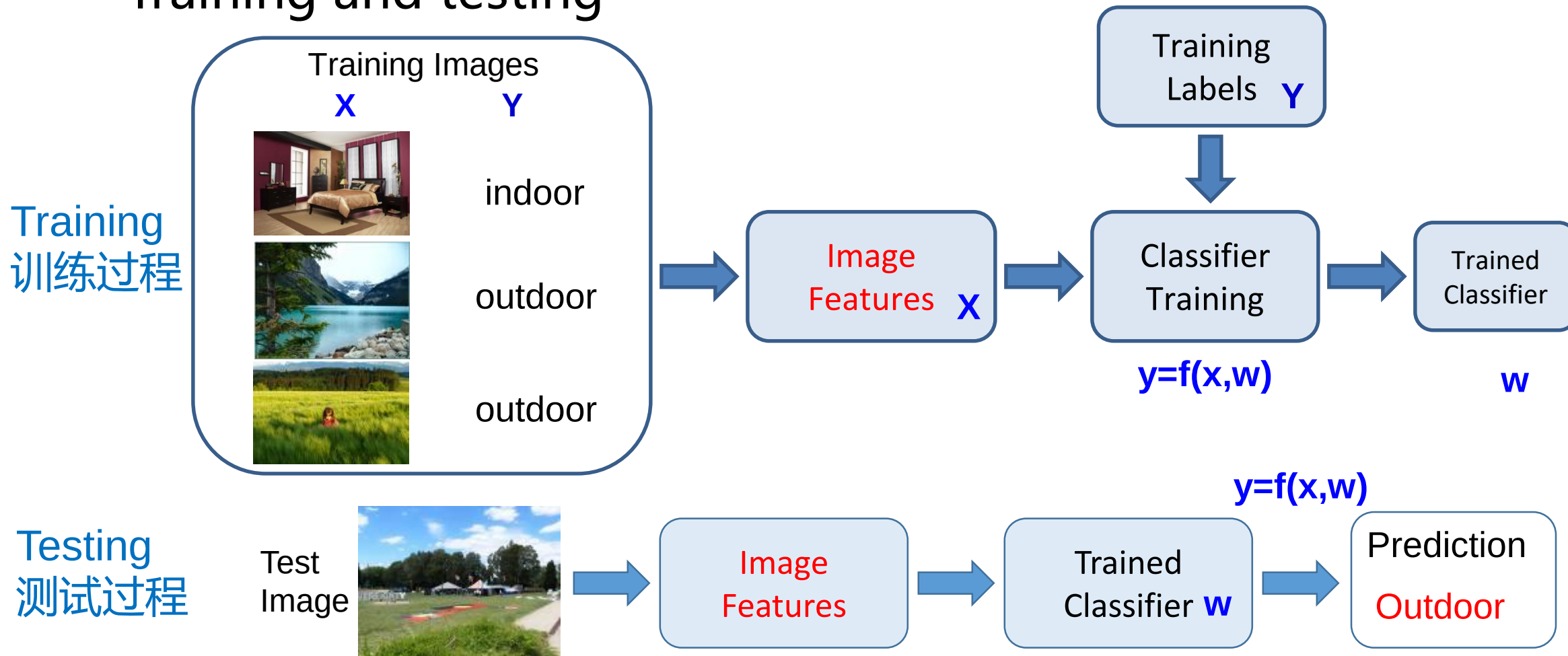


Image Categorization 图像分类



- Training and testing



Training phase as a parameter estimation

- Estimate prediction function $\mathbf{y} = \mathbf{f}(\mathbf{x}, \mathbf{w})$ from labeled data
 - The function has unknown parameters $\mathbf{w}$
 - What is the value of $\mathbf{w}$?

训练过程相当于拟合函数 $\mathbf{y} = \mathbf{f}(\mathbf{x}, \mathbf{w})$ 的最佳参数 $\mathbf{w}$ $\longrightarrow$ 损失函数最小化

- **Goal:** find classifier parameters (weights) $\mathbf{w}$
 - So that $\mathbf{f}(\mathbf{x}^i, \mathbf{w}) = \mathbf{y}^i$ “as much as possible” for all training examples
 - We usually use a **loss function** $L(\mathbf{y}, \mathbf{f})$ to encourage $\mathbf{f}(\mathbf{x}^i, \mathbf{w}) \approx \mathbf{y}^i$

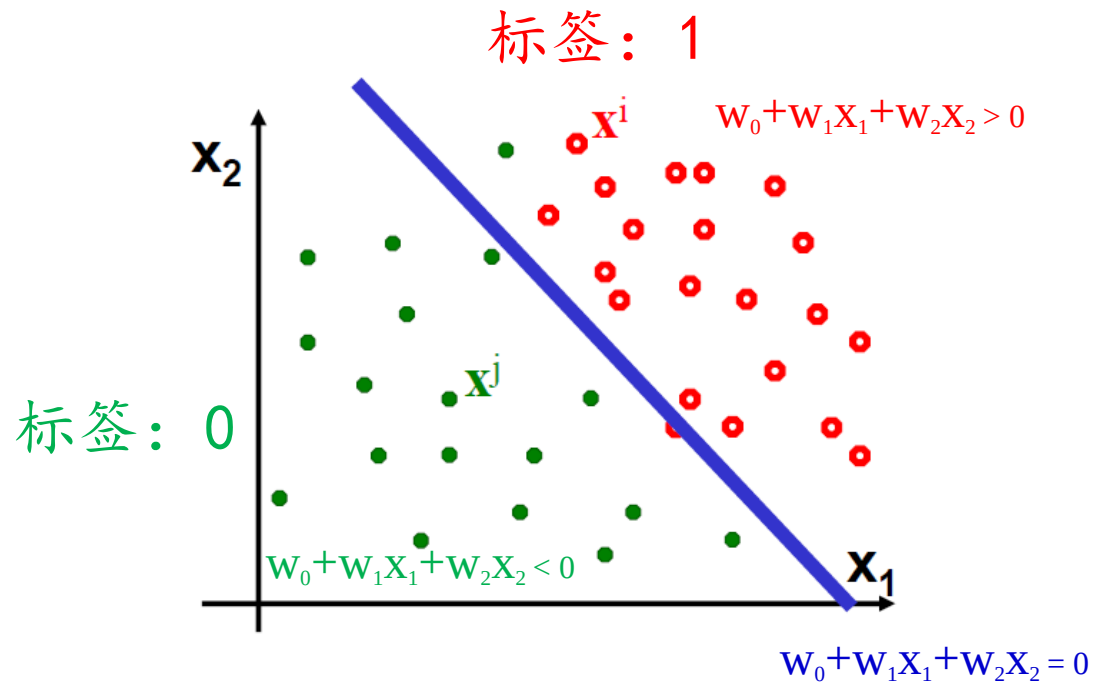
$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \sum_i L(\mathbf{y}^i, \mathbf{f}(\mathbf{x}^i, \mathbf{w}))$$

最优参数 $\swarrow$ $\nwarrow$ 损失函数

Linear classifier 线性分类器



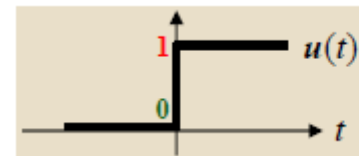
- For two class problem and 2-dimensional data (feature vectors)



Can we find a 2D line to separate the two classes?

$$W_0+W_1X_1+W_2X_2=0 \quad \text{线性分类器}$$

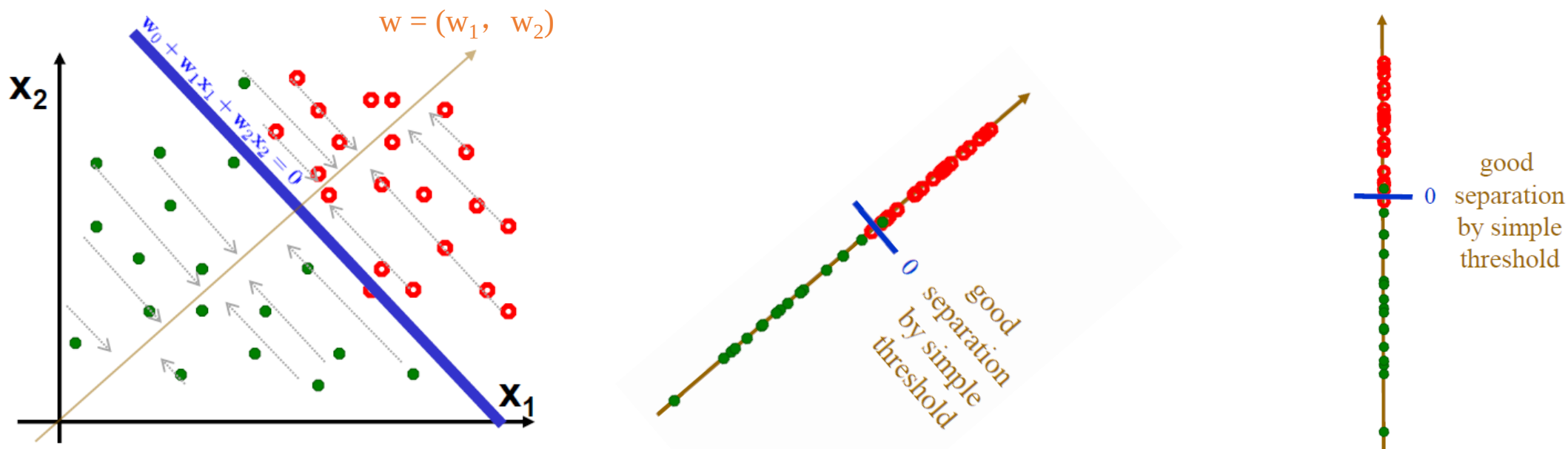
$$f(\mathbf{x}, \mathbf{w}) = u(\mathbf{w}_0+\mathbf{w}_1\mathbf{x}_1+\mathbf{w}_2\mathbf{x}_2) \quad f(\mathbf{x}, \mathbf{w}) \in \{0, 1\}$$



unit step function $u(t) := \begin{cases} 1 & \text{if } t > 0 \\ 0 & \text{O.W.} \end{cases}$
(a.k.a. *Heaviside function*)

Linear classifier 线性分类器

- For two class problem and 2-dimensional data (feature vectors)



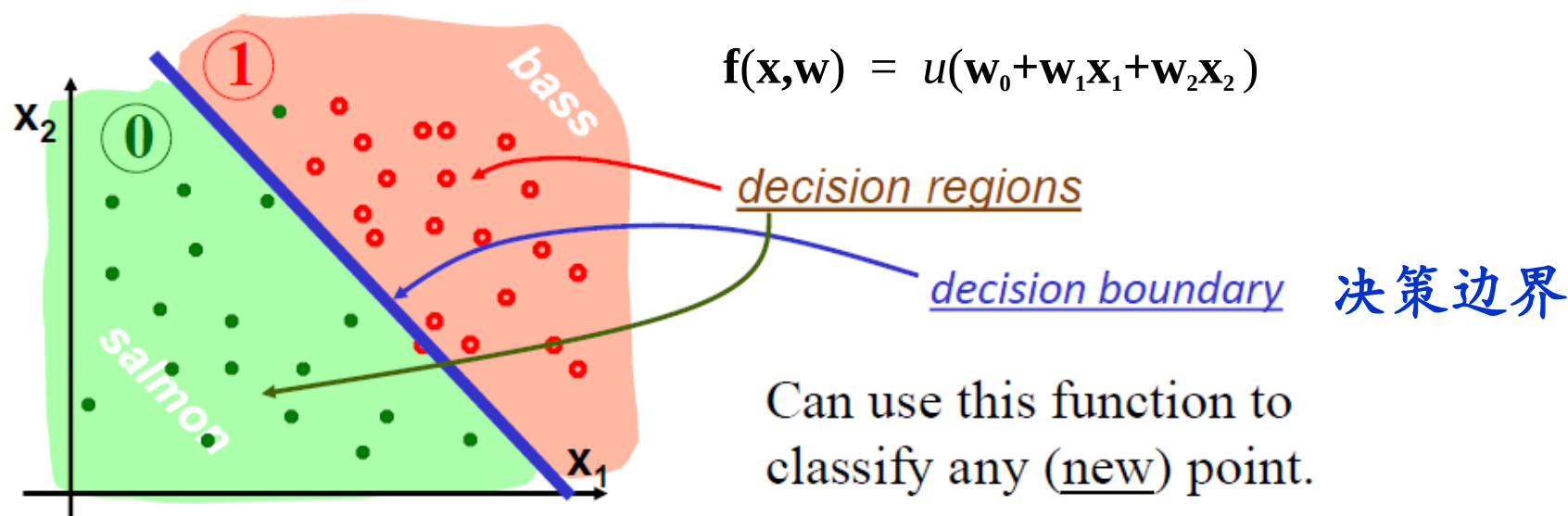
In fact, **any** 2D $\rightarrow$ 1D linear transformation $w = (w_0, w_1, w_2)$ is a **projection onto normal of some hyper-plane**. So, original question really asks if there is a hyper-plane separating data.

对于高维数据，相当于求解一个超平面作为决策边界

Linear classifier 线性分类器



- 二维数据上的二分类问题



可推广到任意维度 m 上的特征向量 $\mathbf{x}$

$$\mathbf{f}(\mathbf{X}, \mathbf{W}) = u(\mathbf{W}^T \mathbf{X})$$

for $\mathbf{W}^T = [\mathbf{w}_0, \mathbf{w}_1, \dots, \mathbf{w}_m]$ and $\mathbf{X}^T = [1, \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m]$

Classifier that makes decisions based on linear combination of features is called a **linear classifier** 线性分类器

Underfitting and overfitting 欠拟合与过拟合



- Underfitting 欠拟合

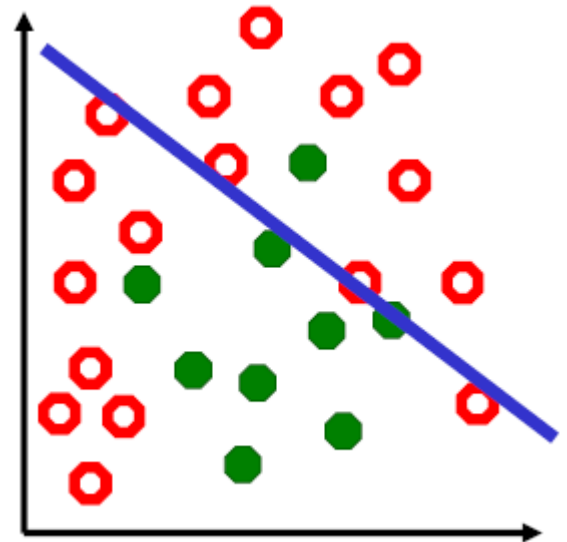
- In some cases, the decision boundary cannot separate the samples well

决策边界不能很好地将样本分开

- Classifier underfits the data

- if it can produce decision boundaries that are too simple for this type of data
 - chosen classifier type (hypothesis space) is not expressive enough

线性分类器



Underfitting and overfitting 欠拟合与过拟合

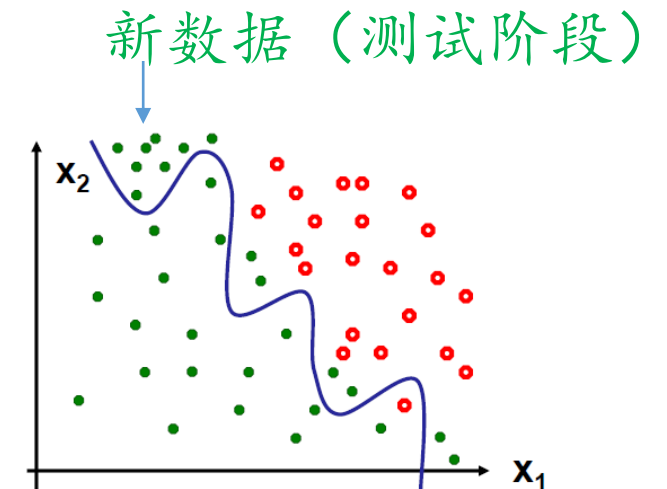
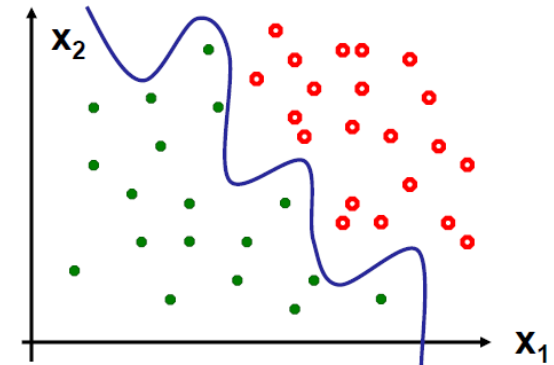


- Overfitting 过拟合

- Model performs perfectly on training data
- But performs bad on testing data

- Reasons

- Amount of data for training is always limited
- Complex model often has too many parameters to fit reliably to limited data
- Complex model may adapt too closely to “random noise”

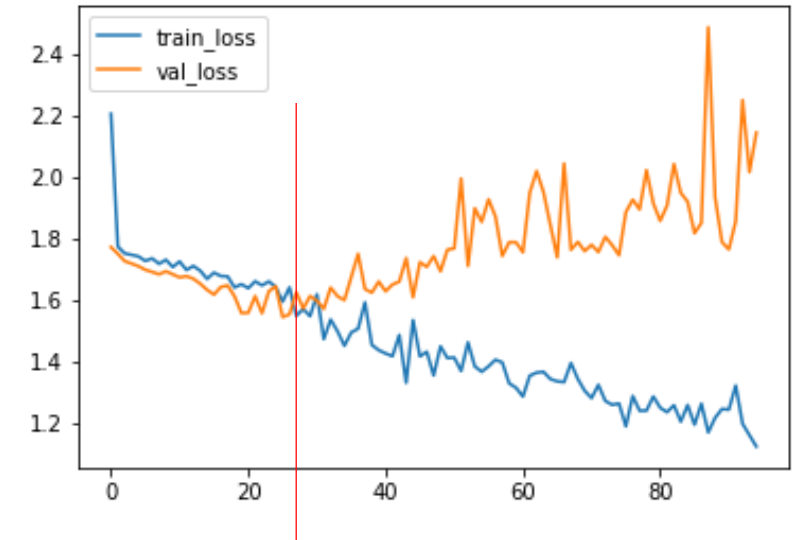
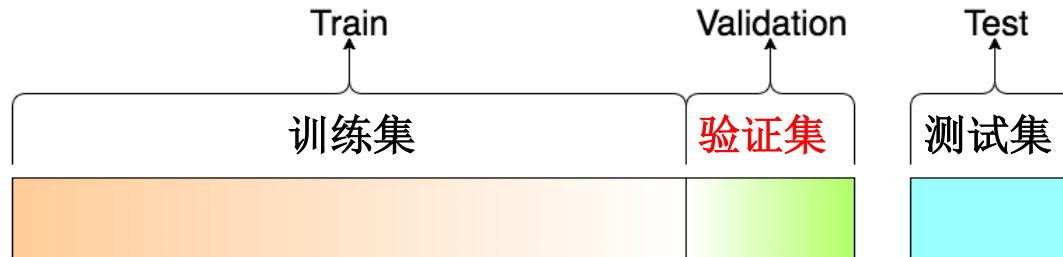


Model Selection: Using Validation Set



How to diagnose overfitting? 如何判断过拟合

- Training error does not provide a good estimate of how well the model will perform on previously unseen data
- Divide training data into two parts:
 - Training set 训练集
 - use for model building
 - Validation set 验证集
 - use for estimating generalization error
- Note: validation set is not the same as test set



Optimal model on
the validation set

Evaluation metrics for classification

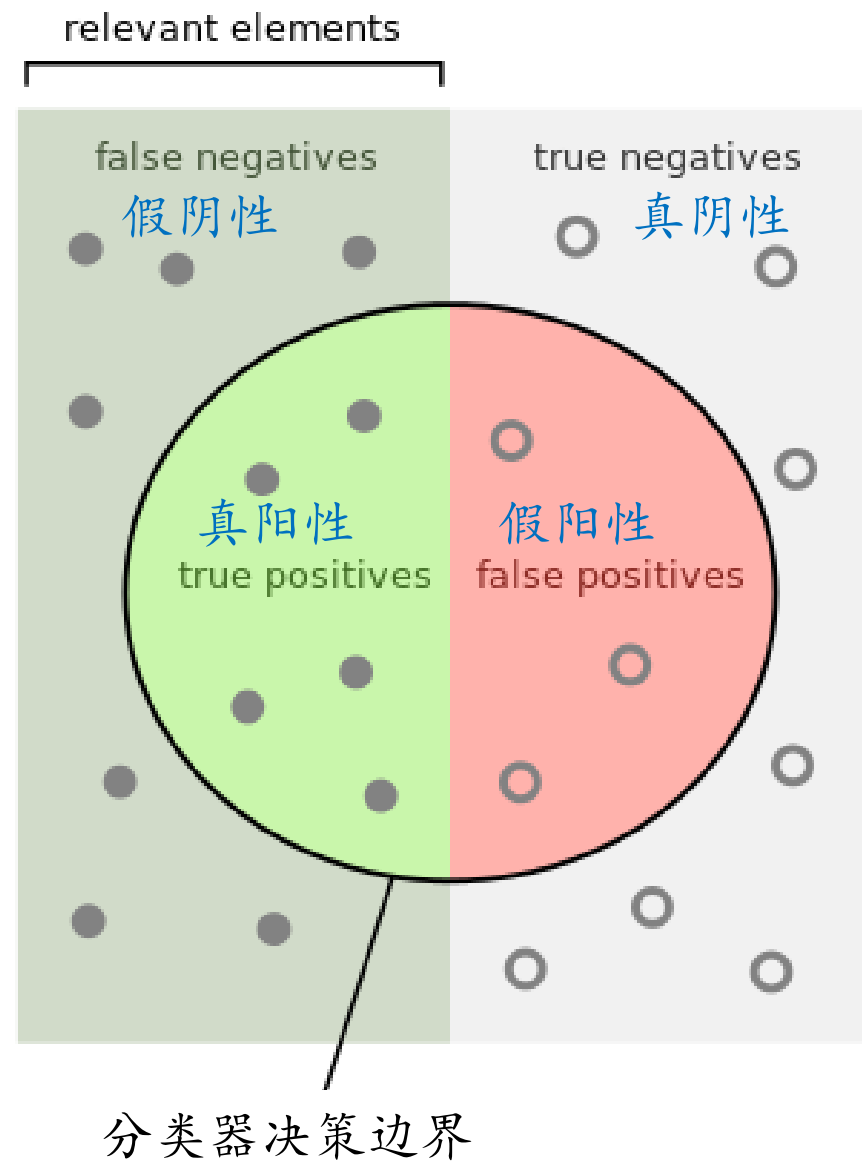
• 分类性能的评价

		Actual	
Predicted		Positives	Negatives
	Positives	TP 真阳性	FP 假阳性
	Negatives	FN 真阴性	TN 假阴性

$$\text{准确率 (Acc)} = \frac{TP+TN}{TP+FP+TN+FN}$$

测试样本中分类正确的样本比例

缺点：在类别不平衡的数据中，准确率可能虚高

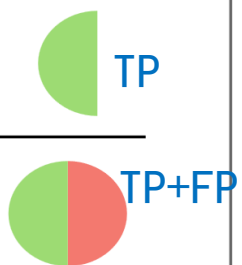


Evaluation metrics for classification

• 分类性能的评价

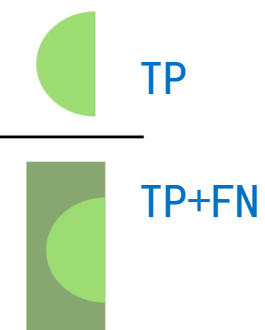
精确度

How many selected items are relevant ?

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$


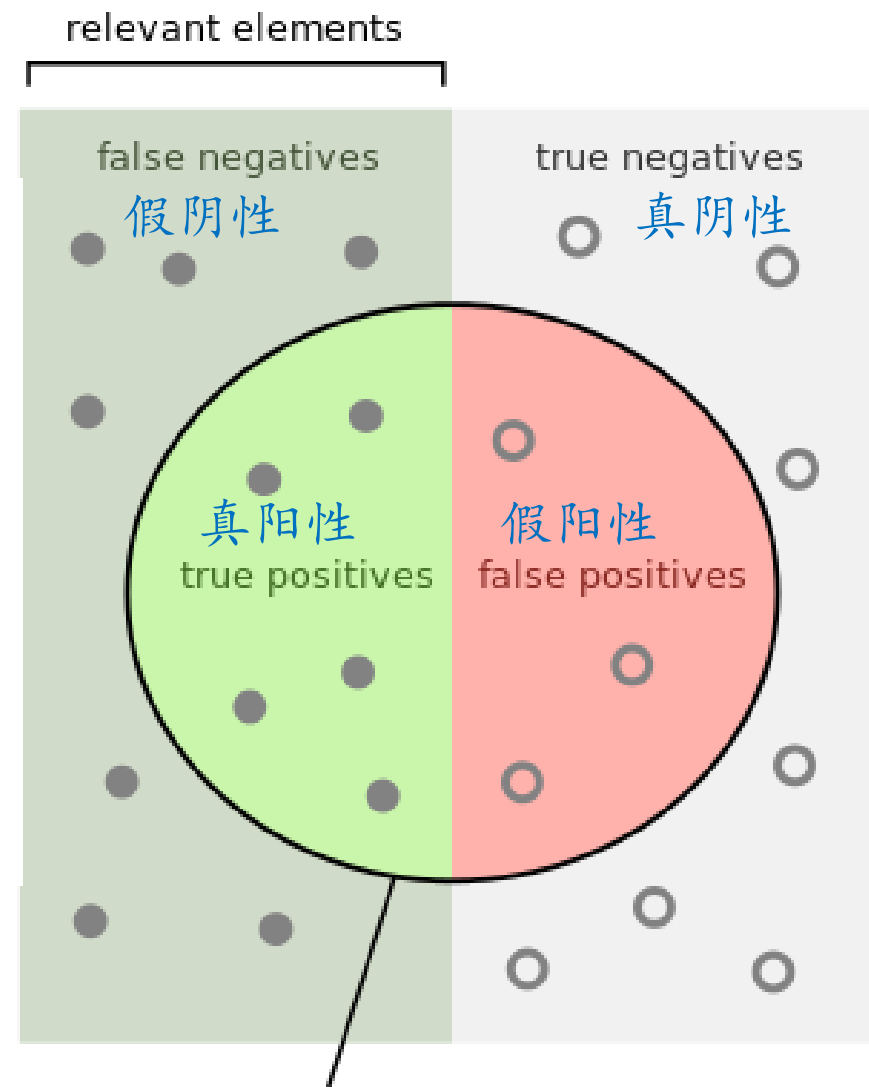
召回率

How many relevant items are selected ?

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$


$$F_1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}$$

F1是精确率和召回率的一个加权平均。



分类器决策边界

Evaluation metrics for classification



- Receiver Operating Characteristic(ROC)

受试者工作特征曲线

- Area Under the Curve (AUC)

曲线下面积 (取值范围0-1)

ROC curve is a graphical representation for displaying the trade off between TPR(**sensitivity**) and FPR(**1-specificity**)

通过对分类器（如逻辑回归）的输出取不同阈值，得到一系列的FP和TP的组合，从而绘制出ROC曲线



Content



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- Bag of features models 特征袋模型
 - Bag of words
 - Spatial pyramid representation
 - Applications

Bag-of-features models 特征袋模型

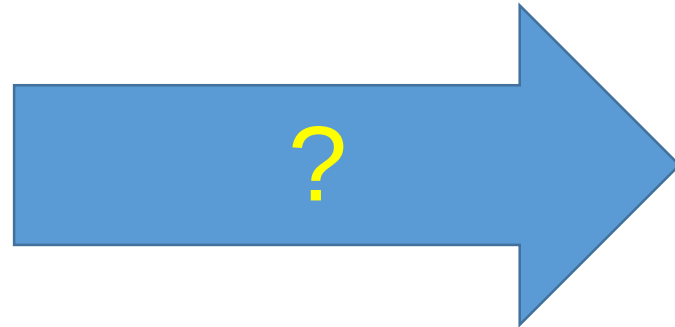
- How to summarize the content of an entire image?
 - In order to classify an image, we need to first transform an image into a feature vector

将图像转化为一个向量（特征）的表示

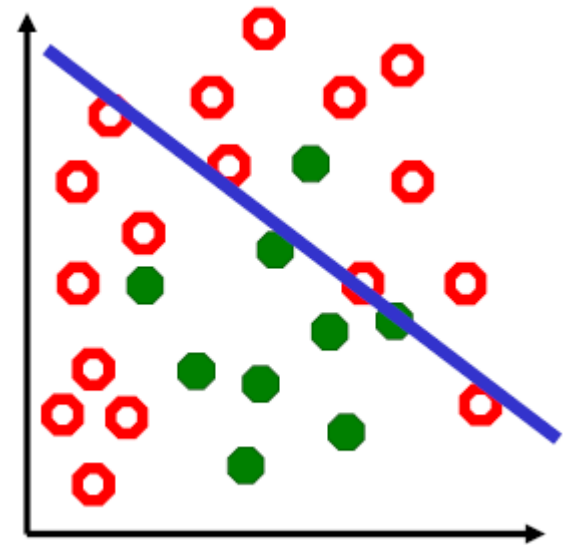
image



Vector/feature

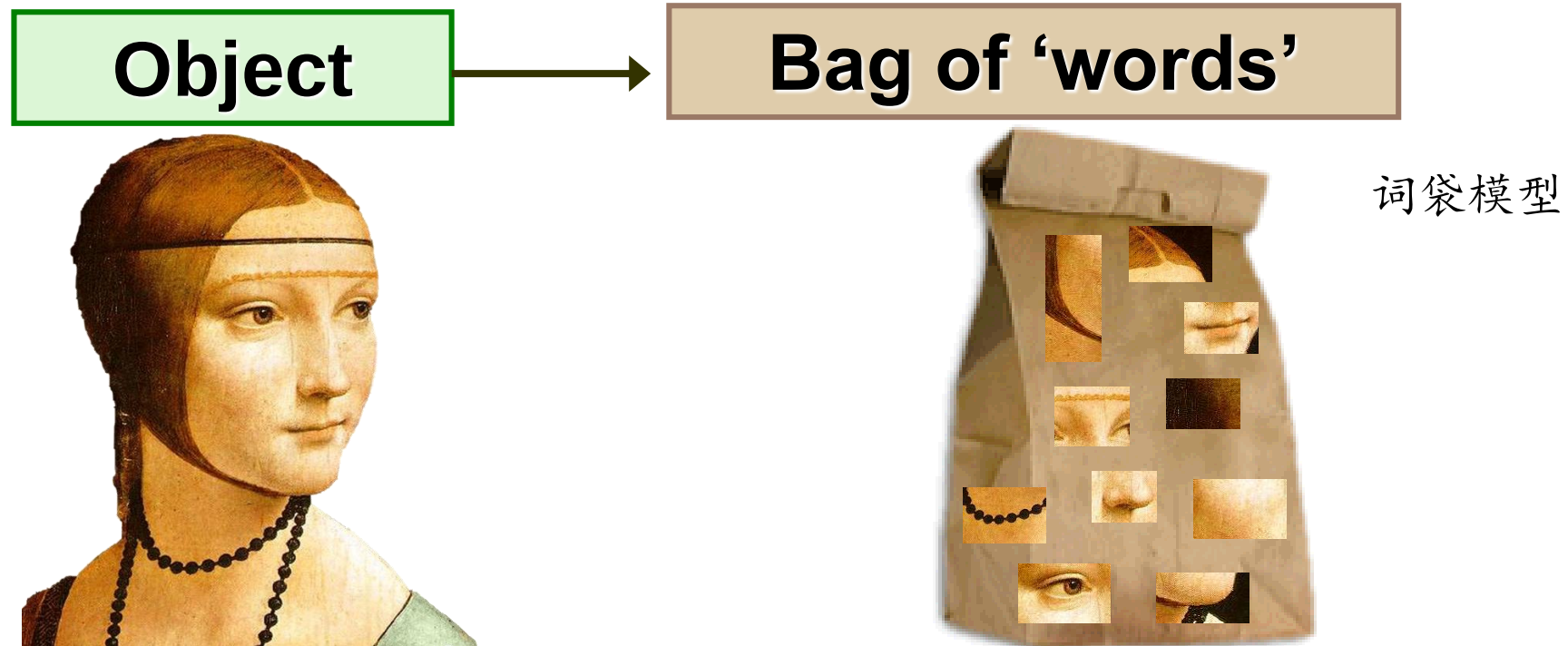


classifier



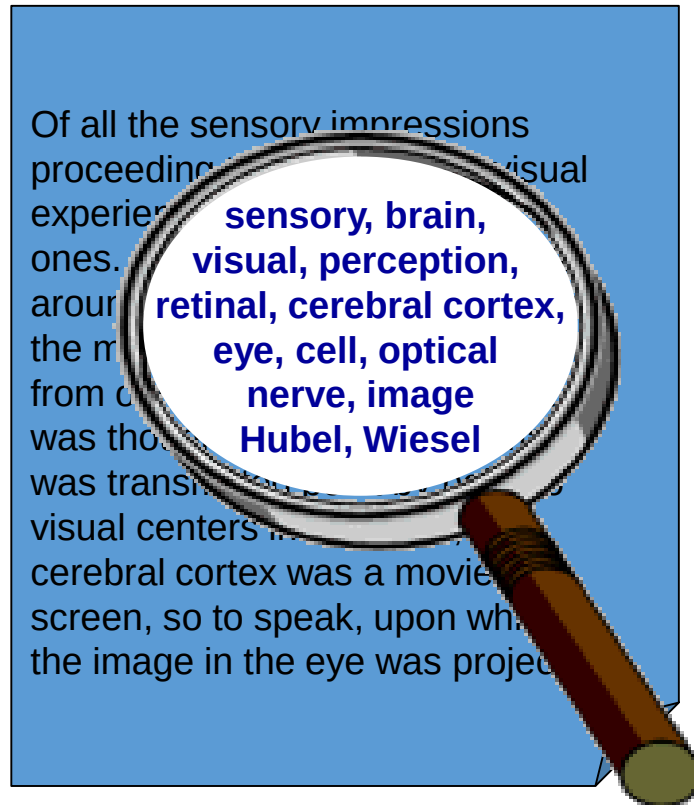
Bag-of-features models 特征袋模型

- How to summarize the content of an entire image?



Bag-of-features models 特征袋模型

- Analogy to documents 文本分类



Bag-of-features models 特征袋模型



Origin 1: Bag-of-words models (词袋模型)

- Orderless document representation: frequencies of words from a dictionary

一段文字



单词的出现频率统计



长度固定的一个特征向量

“ The most famous team lost the ball game, but the score of that team was high.”

```
BoW =  
{ "team":2, "ball":1, "game":1, "season":0, "win":0,  
  "play":0, "score":1};
```



[2, 1, 1, 0, 0, 0, 1]

Bag-of-features models 特征袋模型



Origin 1: Bag-of-words models

- A document becomes a 'term/word' vector
- Each term is a component (attribute) of the vector
- The value is the number of times the word is used

文本转化为
特征向量表示

	team	income	game	currency	score	ball	weight	drug	food	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

Application:

News classification

- Economy
- Sports
- Healthcare
- ...

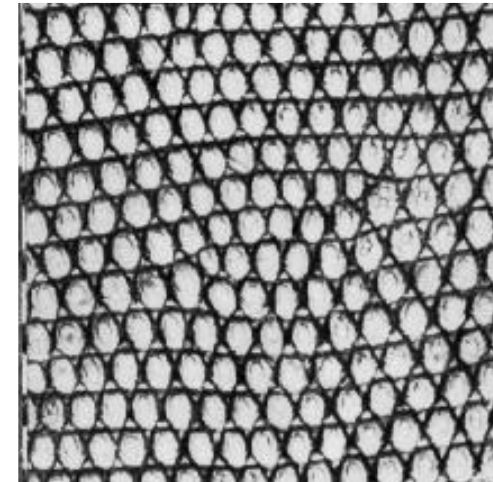
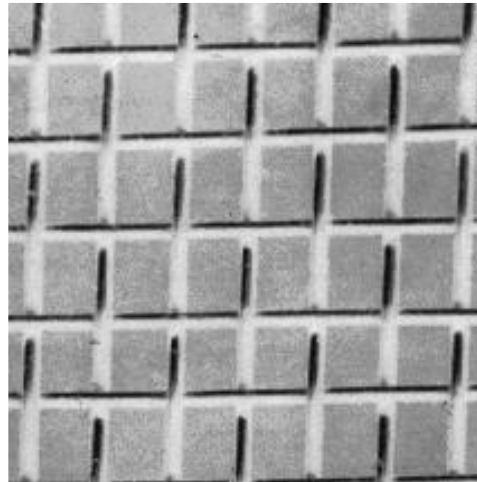
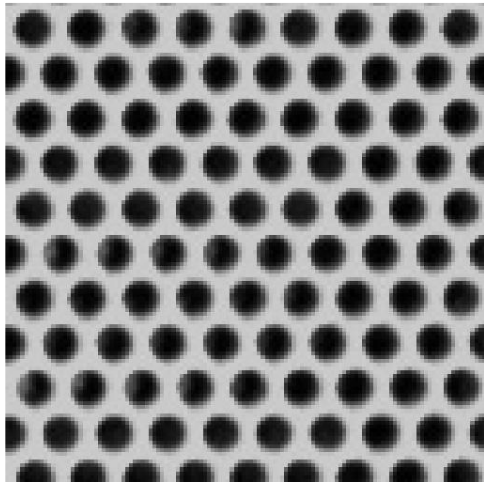
Bag-of-features models 特征袋模型



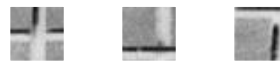
Origin 2: Texture recognition 纹理识别

- Characterized by repetition of basic elements or *textons* 纹理单元
- For stochastic textures, the identity of textons matters, not their spatial arrangement

图像



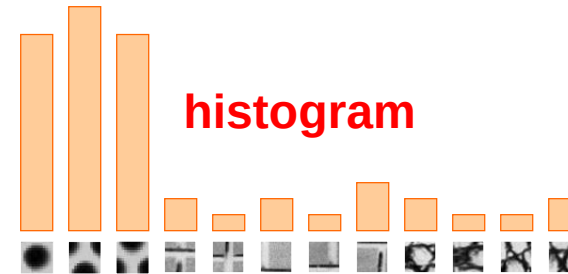
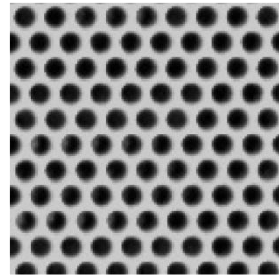
纹理单元



Bag-of-features models 特征袋模型

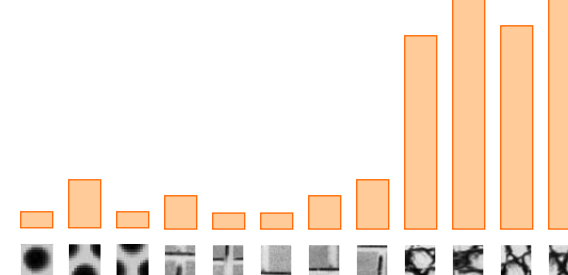
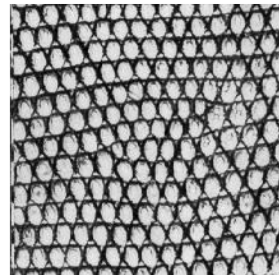
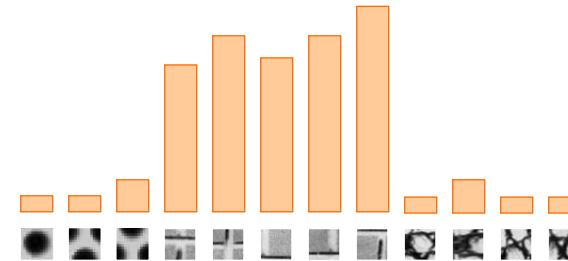
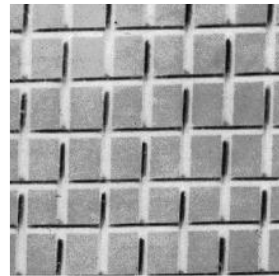


Origin 2: Texture recognition 纹理识别



histogram

Universal texton dictionary



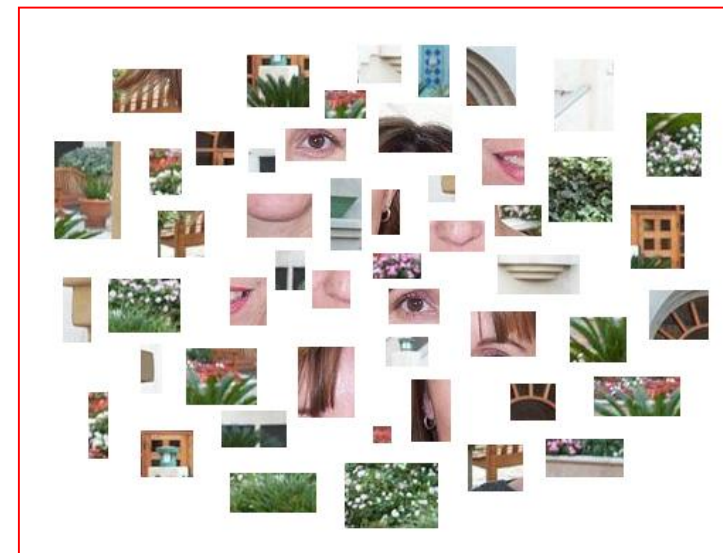
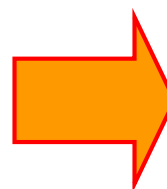
不同图像的纹理
单元直方图

Bag-of-features models 特征袋模型

- Image representation 图像表示



Image

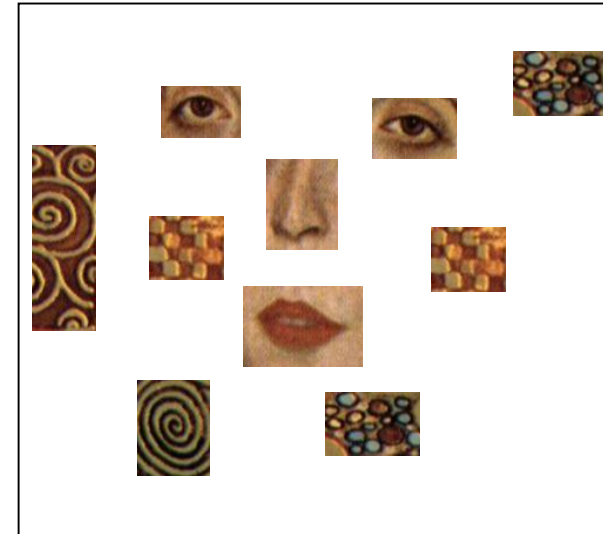
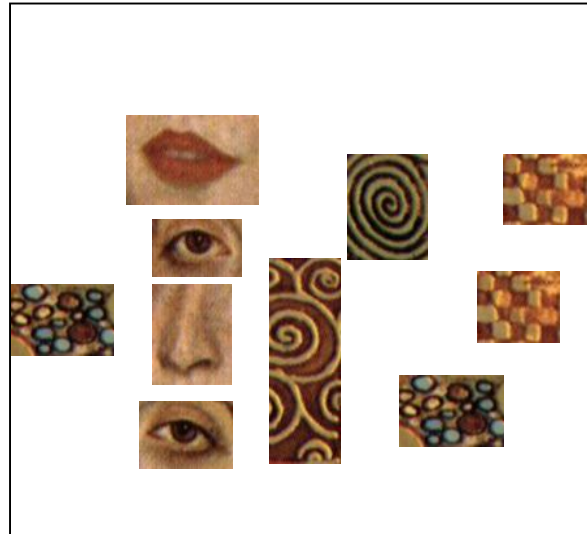
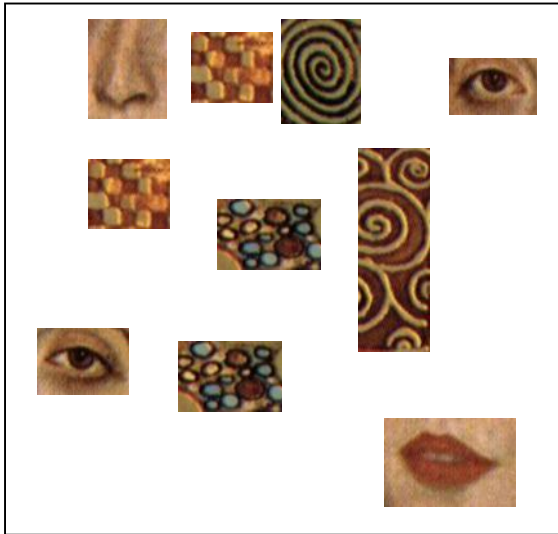


A bag of features
(dictionary)
特征字典

Bag-of-features models 特征袋模型

- All of these are treated as being the same
 - The same dictionary with different orders

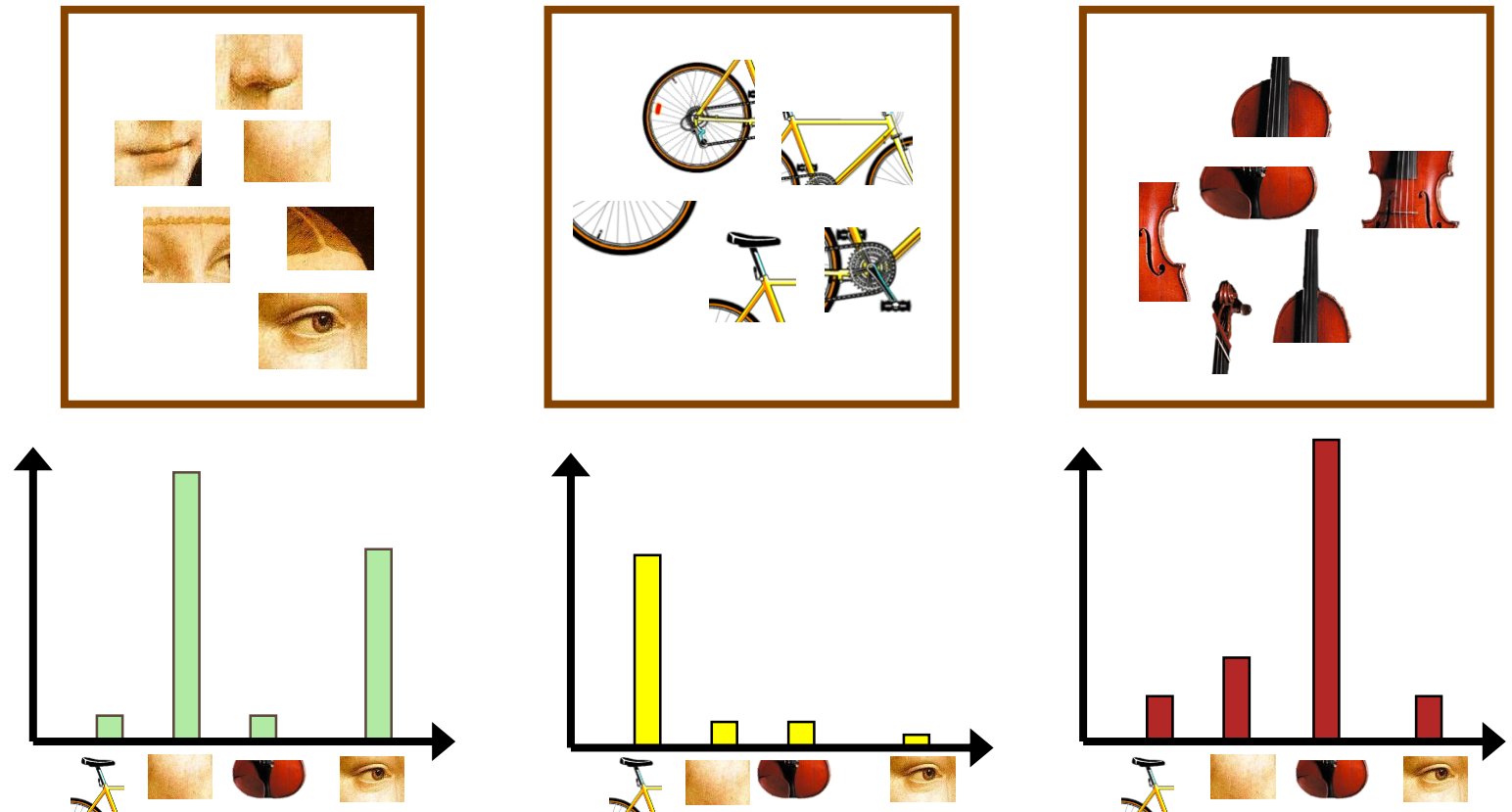
特征字典：与顺序无关



Bag-of-features models 特征袋模型

- Bag-of-features steps 特征袋的构建过程

- 1, Feature extraction
- 2, Learn “visual vocabulary”
- 3, Quantize features using visual vocabulary
- 4, Represent images by frequencies of “visual words”



Bag-of-features models 特征袋模型



1. Feature extraction 特征提取

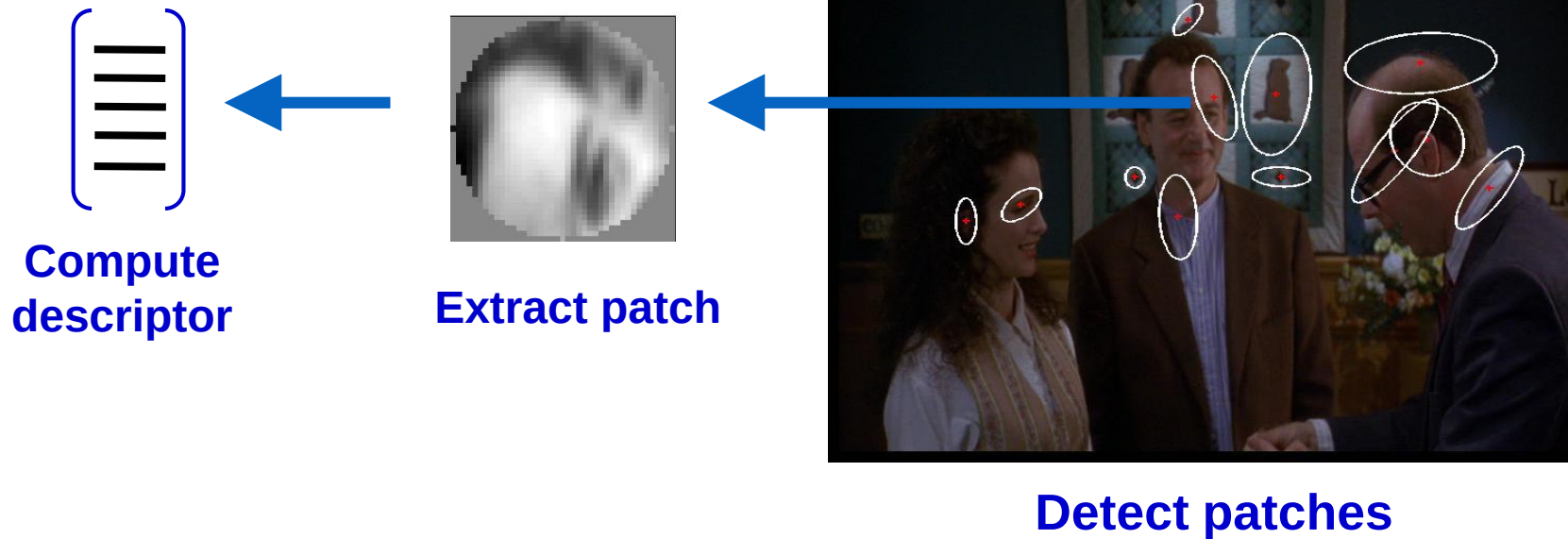
- Regular grid or interest regions



Bag-of-features models 特征袋模型

1. Feature extraction 特征提取

- Regular grid or interest regions

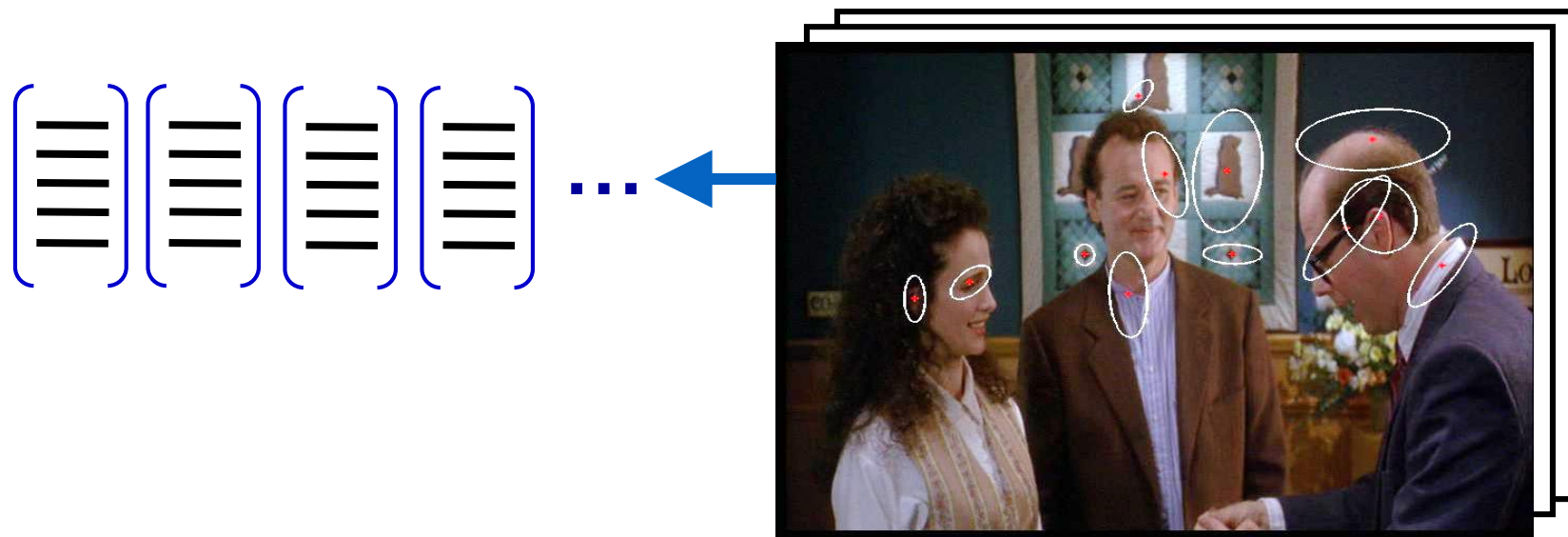


Bag-of-features models 特征袋模型



1. Feature extraction 特征提取

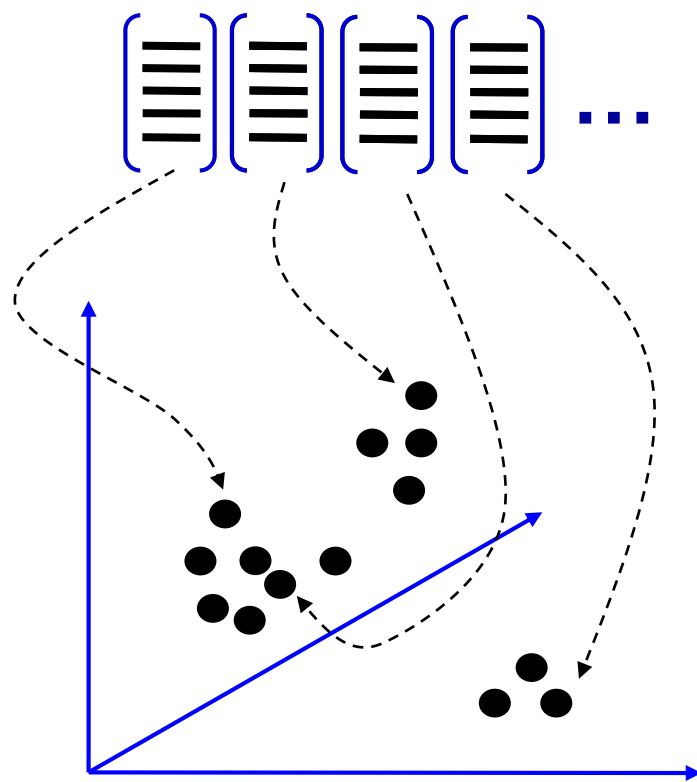
- Regular grid or interest regions



Bag-of-features models 特征袋模型



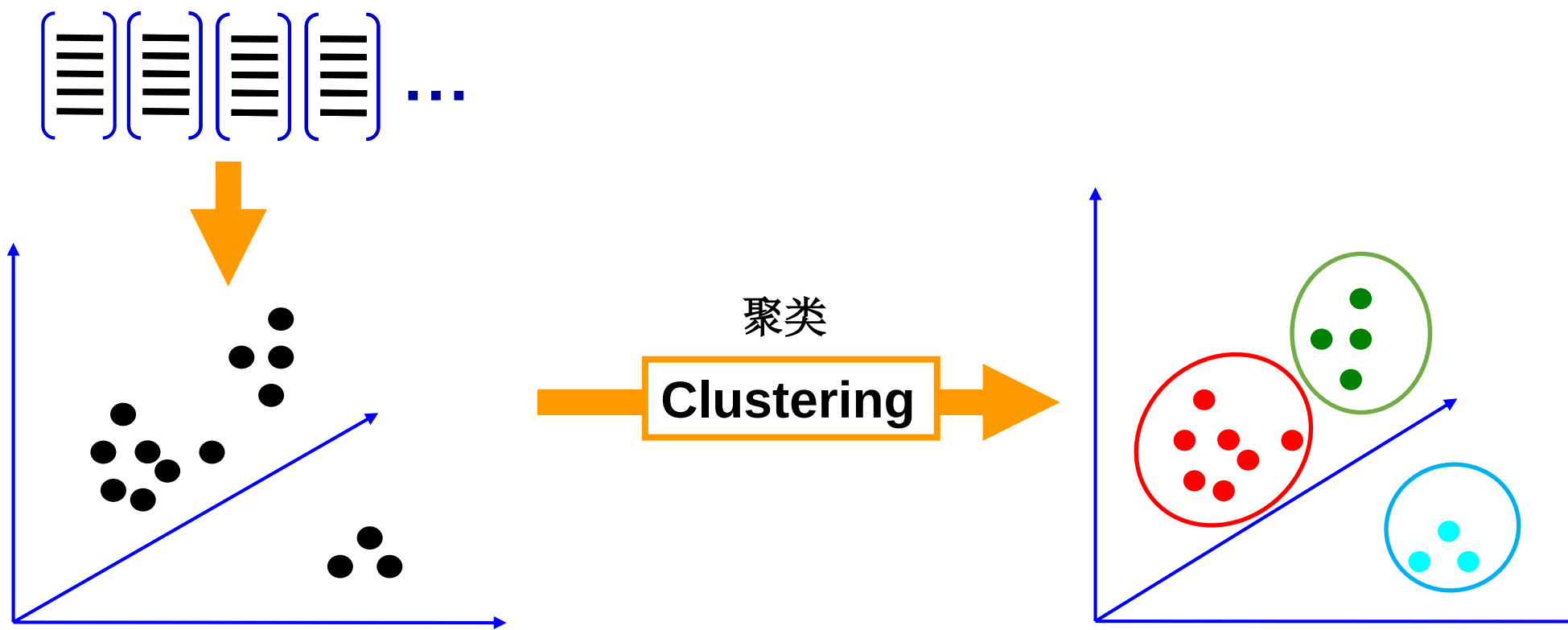
2. Learning the visual vocabulary 视觉词汇（字典）的学习



Bag-of-features models 特征袋模型



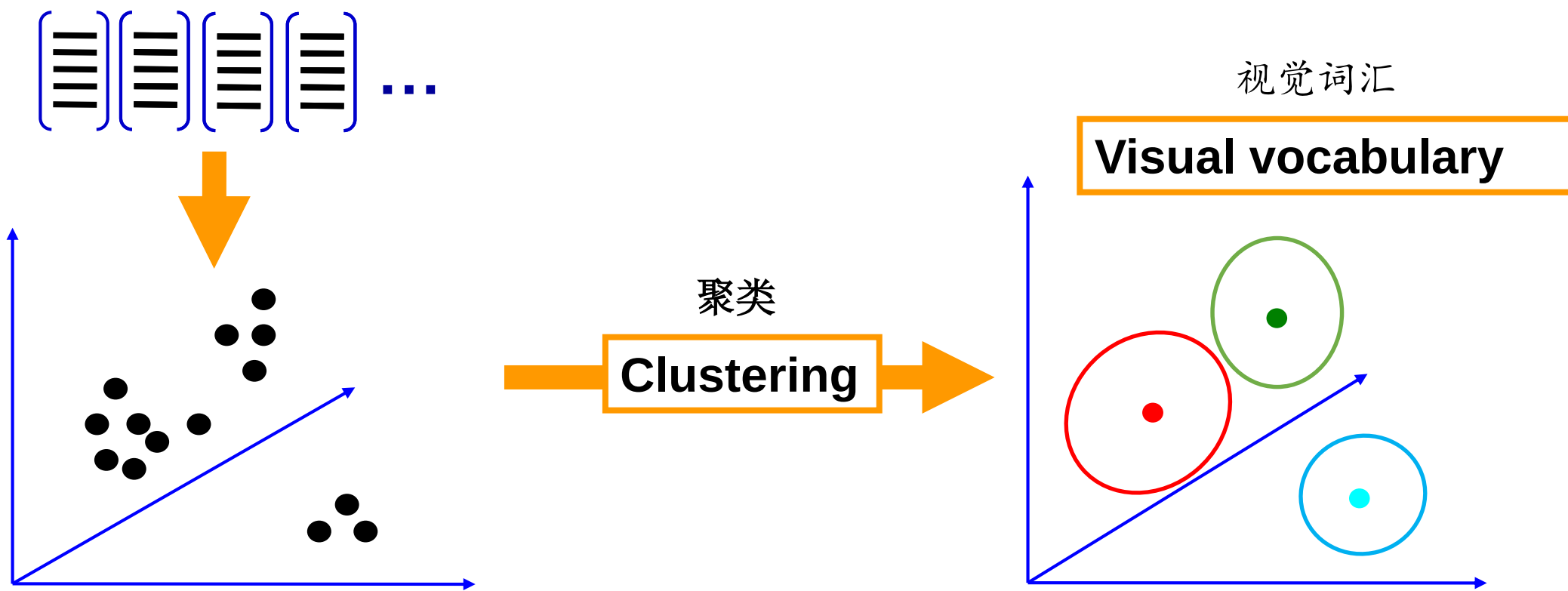
2. Learning the visual vocabulary 视觉词汇（字典）的学习



Bag-of-features models 特征袋模型



3. Quantize the visual vocabulary 视觉词汇（字典）的数值化



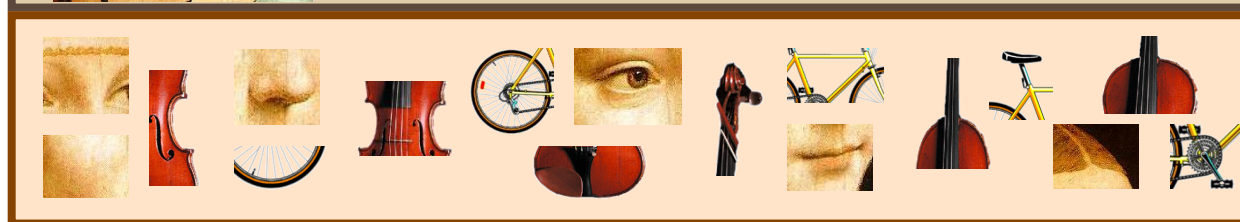
Bag-of-features models 特征袋模型

4, Represent images by frequencies of “visual words”

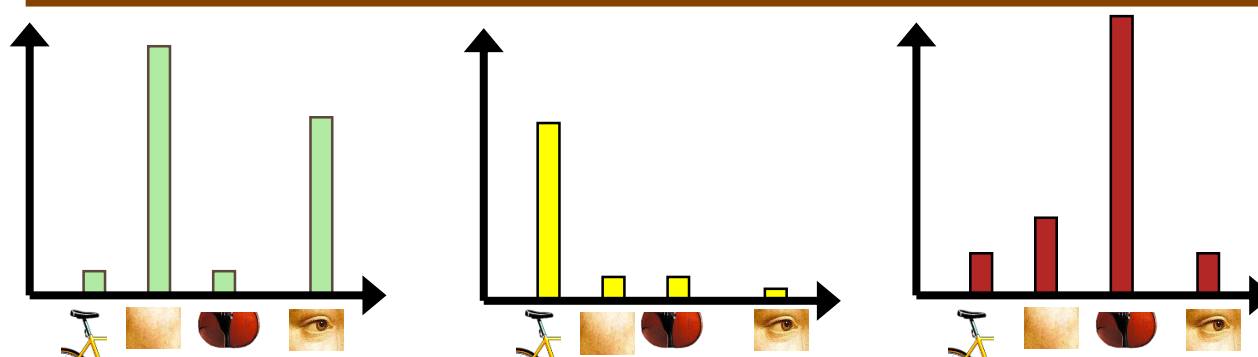
images



Visual words



Bag of visual words
histograms



将图像表达
为视觉词汇
的直方图（
向量）

Bag-of-features models 特征袋模型

Example real codebook



Bag-of-features models 特征袋模型



优点

- + flexible to geometry / deformations / viewpoint
- + compact summary of image content 对图像的几何变化、视角等不敏感
- + provides fixed dimensional vector representation for sets
- + very good results in practice

缺点

- background and foreground mixed when bag covers whole image -> *is it really instance recognition?*
- optimal vocabulary formation remains unclear
- basic model ignores geometry – must verify afterwards, or encode via features

忽略了图像中物体的位置关系（结构信息）

Bag-of-features models 特征袋模型

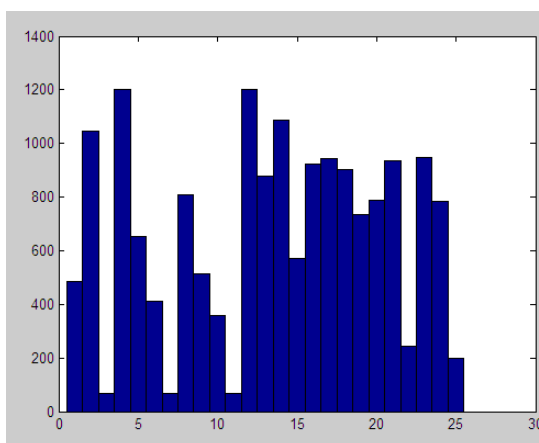


But what about layout?

All of these images have the same color histogram.
How to extend bag of words?



特征袋模型忽略了图像中物体的位置关系（结构信息）

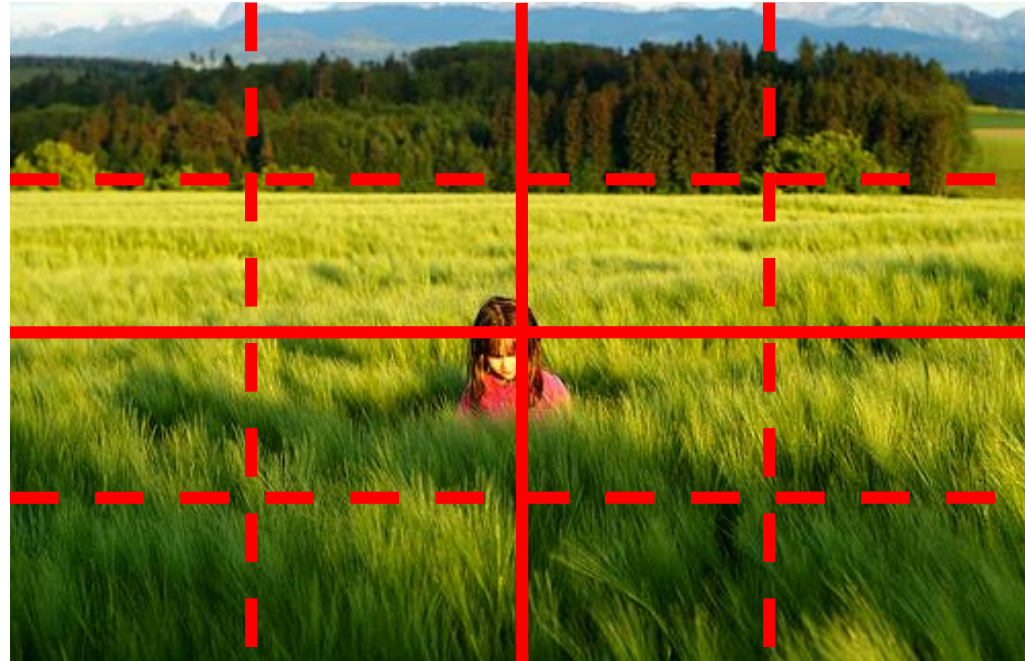


Bag-of-features models 特征袋模型



Spatial pyramid 空间金字塔

Compute histogram in
each spatial bin



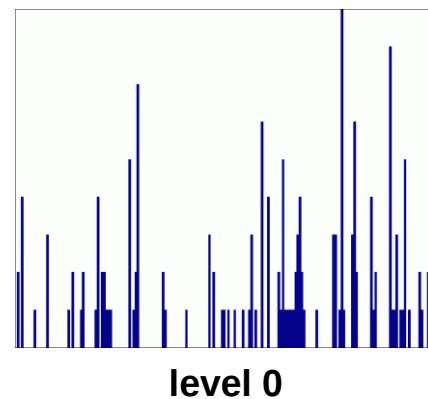
Bag-of-features models 特征袋模型

Spatial pyramid representation

- Extension of a bag of features
- Locally orderless representation at several levels of resolution



空间金字塔



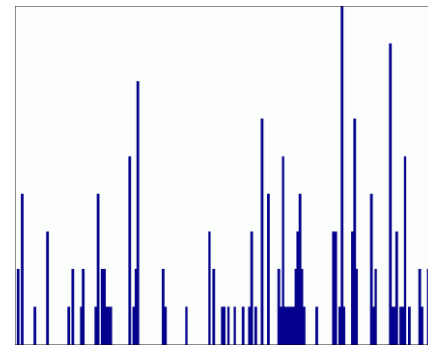
Bag-of-features models 特征袋模型

Spatial pyramid representation

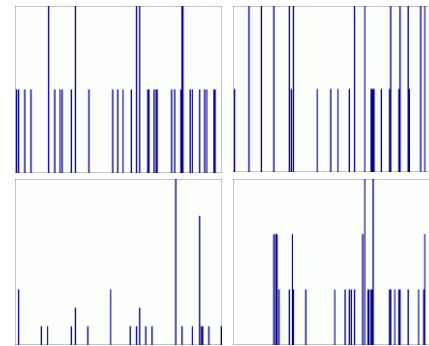
- Extension of a bag of features
- Locally orderless representation at several levels of resolution



空间金字塔



level 0

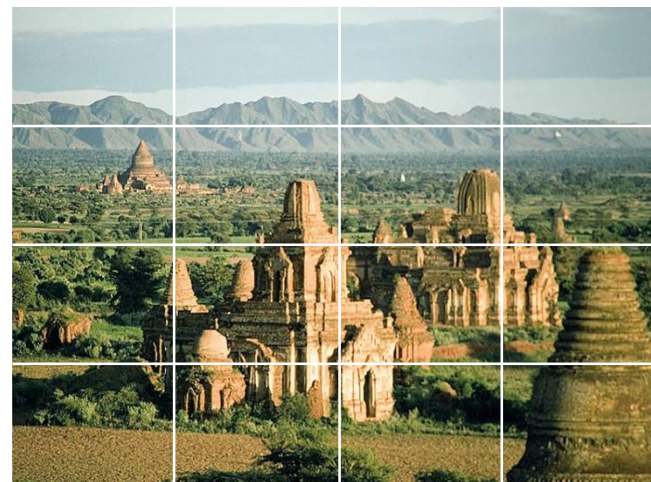


level 1

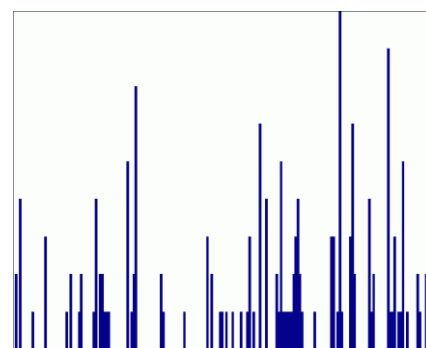
Bag-of-features models 特征袋模型

Spatial pyramid representation

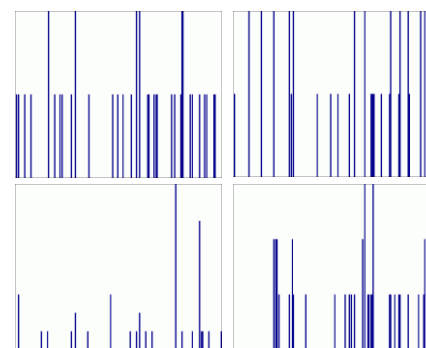
- Extension of a bag of features
- Locally orderless representation at several levels of resolution



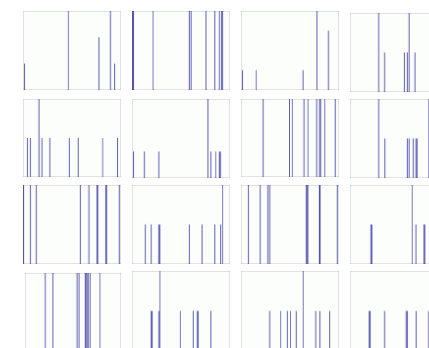
空间金字塔



level 0



level 1

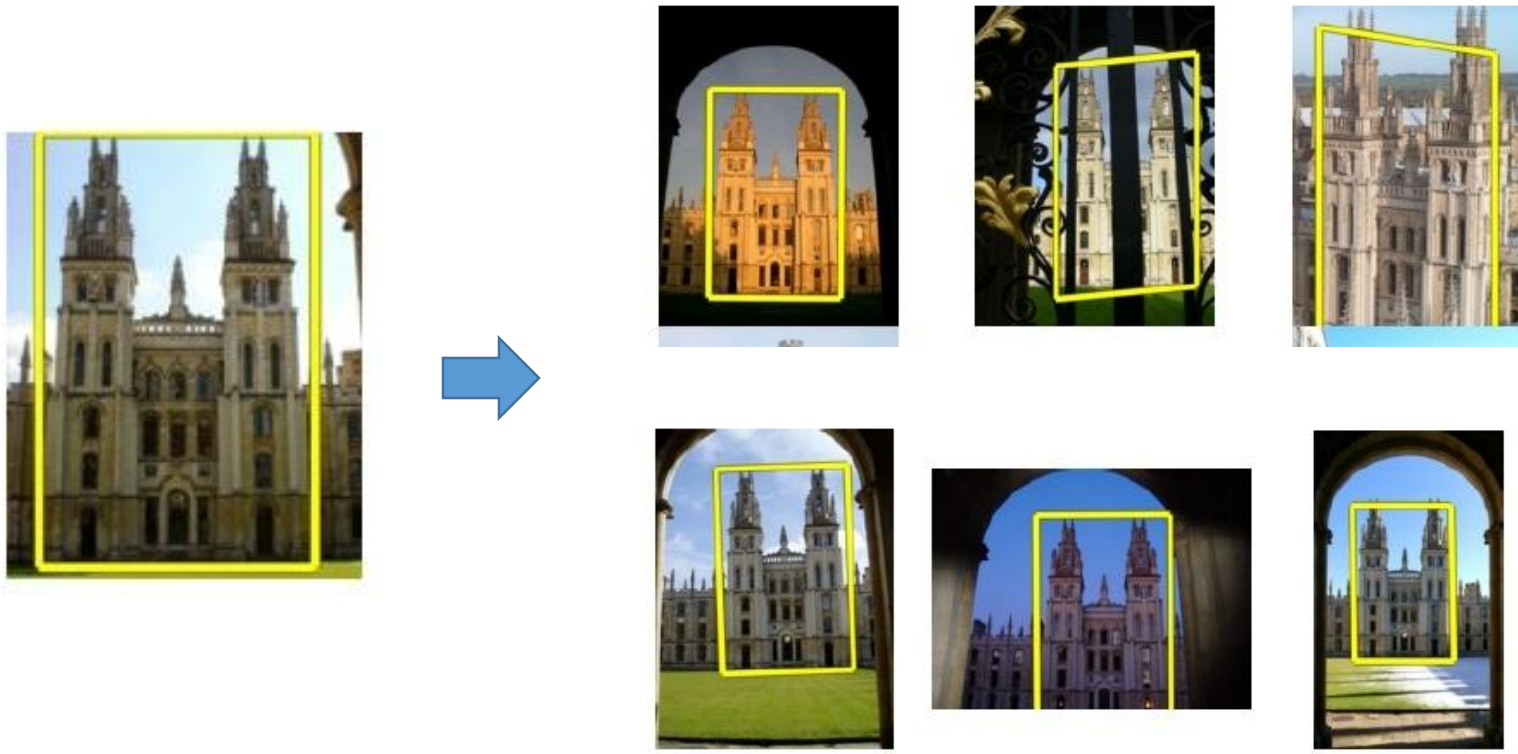


level 2

Bag-of-features models 特征袋模型

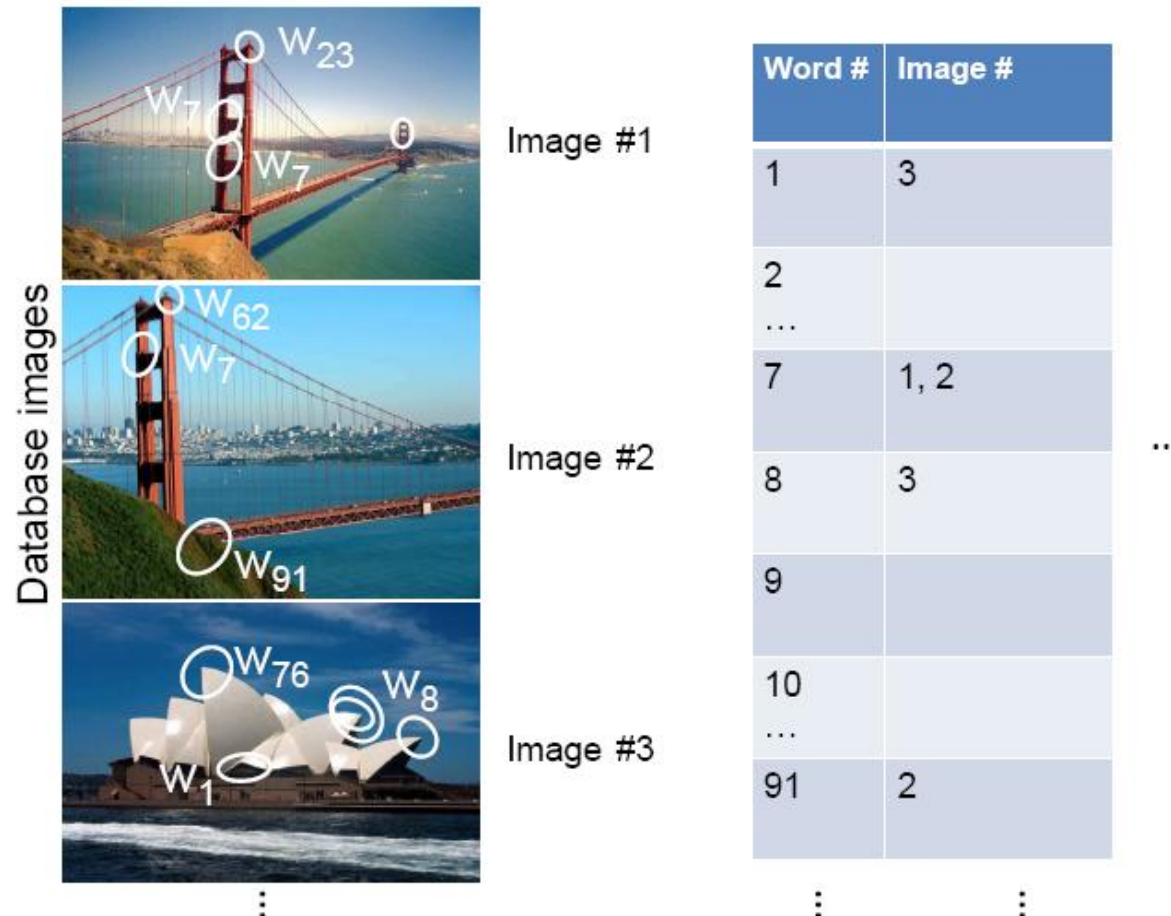
Application: Image search 图像搜索

- How can we quickly find images in a large database that match a given image region?



Bag-of-features models 特征袋模型

Build Inverted Index from Database



Bag-of-features models 特征袋模型

Query Inverted Index



New query image

Word #	Image #
1	3
2	
...	
7	1, 2
8	3
9	
10	
...	
91	2
⋮	⋮

Candidate matches

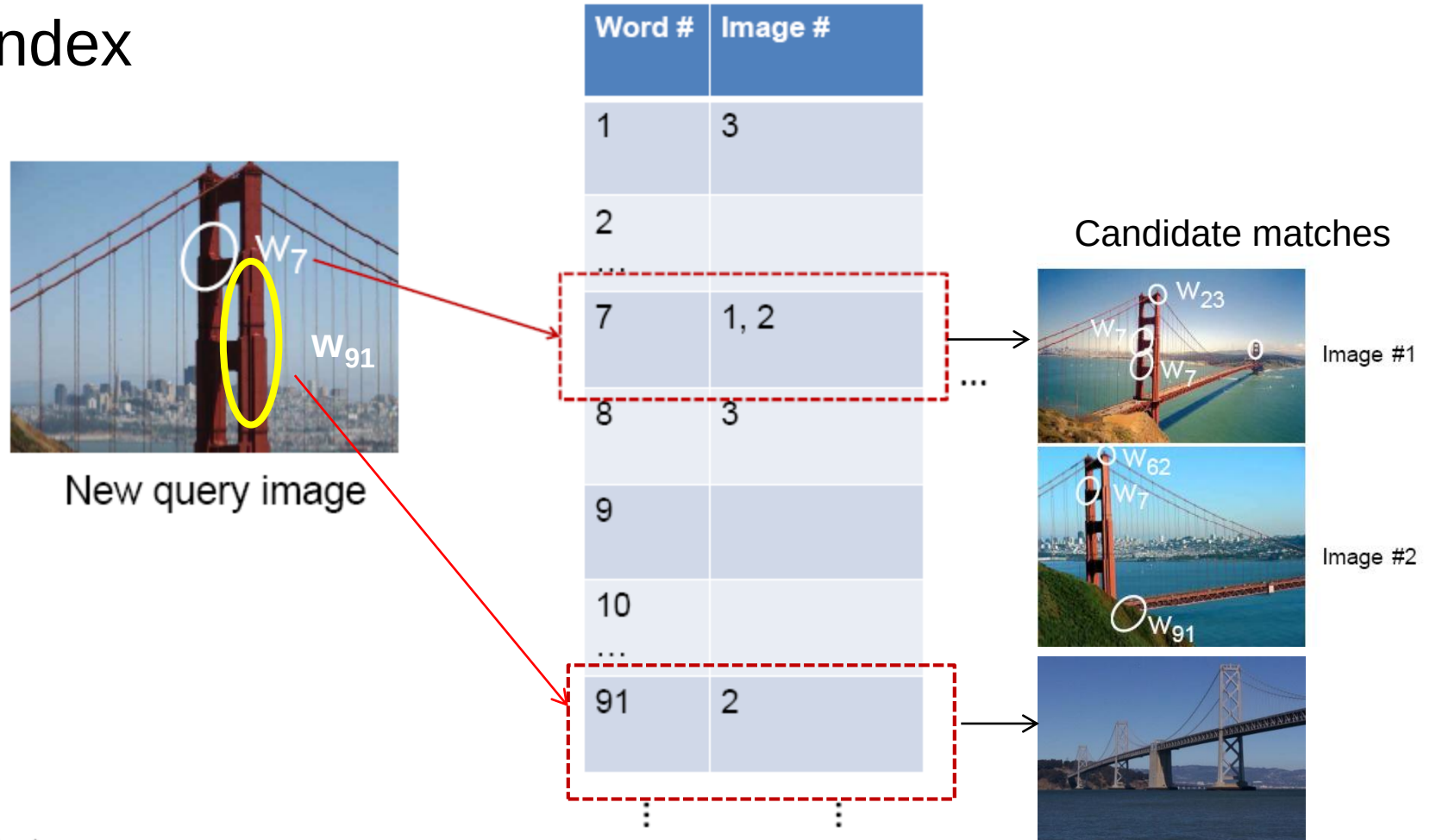


Image #1

Image #2

Bag-of-features models 特征袋模型

Query Inverted Index



小结

• 监督学习

- 线性与非线性分类器
- 过拟合与欠拟合
- 特征袋模型
 - Bag of words
 - 空间金字塔表达

$$y = f(x, w)$$

